INCH-POUND

MIL-M-38510/313A
AMENDMENT 2
30 December 1999
SUPERSEDING
AMENDMENT 1
8 December 1987

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW POWER SCHOTTKY TTL,
SCHMITT-TRIGGER POSITIVE-NAND GATES
AND INVERTERS, MONOLITHIC SILICON

Inactive for new design after 18 April 1997

This amendment forms a part of MIL-M-38510/313A, dated 22 July 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.2.3, outline letter C-2: Delete "X" and substitute "2".

1.2.3, add the following case outline:

"X C-2A (20-terminal, .350" x .350"), square chip carrier package".

1.3, thermal resistance, junction-to-case (θ_{JC}): delete and substitute the following:

"Thermal resistance, junction-to-case (θ_{JC}):
Cases A, B, C, D, X and 2 - - - - - - - - - (See MIL-M-38510, appendix C)".

1.3 junction temperature (T_J), delete and substitute:

"Juction temperature (T_J) 2/ - - - - - - - - - +175°C.

Following footnote 1/ add footnote 2/ as follows:

"2/ Maximum junction temperature shall not be exceeded for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883."

PAGE 2

3.2.3, delete in its entirety and substitute the following:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

The attached insertable pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
9	REPRINTED WITHOUT CHANGE
10	10

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PAGE 3

* TABLE I: In test column, add "2/" after Low-level input current.

* TABLE I: min limit for Low-level input current, change "-.17" to "-.12".

* TABLE I: max limit for Low-level input current, change "-.41" to "-.36".

PAGE 4

3.6, second sentence: Delete in its entirety.

TABLE II: Delete "(preburn-in)".

TABLEII: Group A test requirements, class B devices, add "10, 11".

TABLE II: "Additional electrical subgroups for group C periodic inspections", delete in its entirety.

* After footnote 1/ at end of TABLE I, add footnote 2/ as follows:

* "2/ For CKT E, device type 03, I_{IL} limits are -0.17 mA min and -0.41 mA max."

PAGE 5

4.2 c Delete the first sentence and substitute the following:

"c. The percent defective allowable PDA shall be specified in MIL-M-38510."

4.4.3b, delete in its entirety.

PAGE 6

FIGURE 1, device type 01: delete "Case X" and substitute "Cases 2, X" make these same changes to device types 02 and 03.

PAGES 13 THROUGH 18

TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 14

TABLE III, footnote 2, delete footnote 2, delete "Case X" and substitute "Case X, 2".

Make these same changes on pages 16 and 18.

PAGE 19

4.4.3c (3), delete and substitute:

"(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

6.1: Delete in its entirety.

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The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5962-1862)

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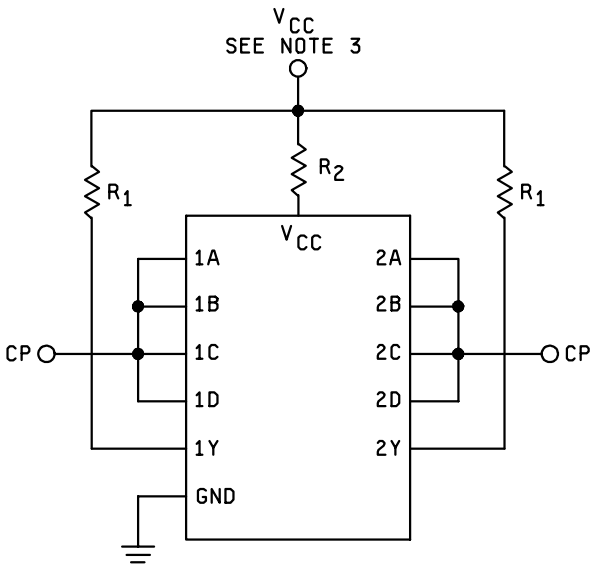
Device type 01



NOTES:
1. CP = 100 kHz ± 50 percent square wave; duty cycle = 50 percent ± 15 percent; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.
2. R1 = 1 kΩ ± 5 percent.
3. V_{CC} and R_2 shall be chosen to insure 5.0 V at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuits.

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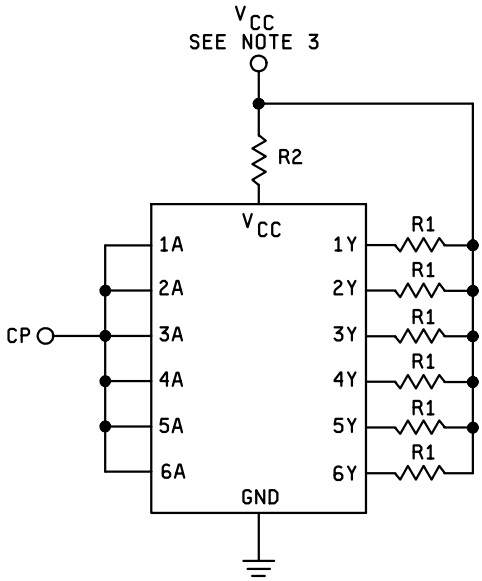
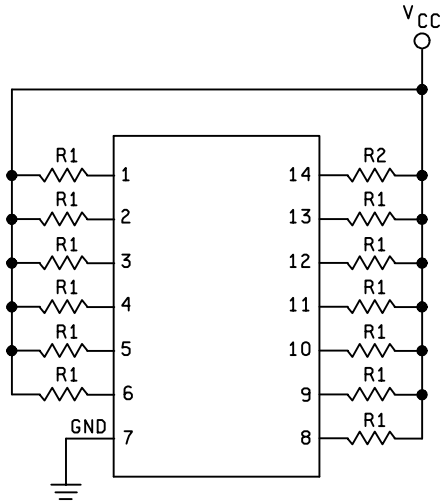
Device type 02



NOTES:
2. CP = 100 kHz ± 50 percent square wave; duty cycle = 50 percent ± 15 percent; V_{IL} = 0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.
2. R1 = 1 kΩ ± 5 percent.
3. V_{CC} and R2 shall be chosen to insure 5.0 V at device V_{CC} terminal.



NOTES:
1. High temperature (175°C), condition F, burn-in circuit only.
2. R1 = 2 kΩ.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuits – Continued.

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